

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester I Examination, 2016-17

Paper No. MTB-102: Geometry

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Find the latus rectum and the point on the conic $\frac{l}{r} = 3 - 8 \cos \theta$ whose radius vector is 2.
- (b) Find the length of the focal chord of a parabola which passes through a point whose vectorial angle is α .
- (c) Find the intercepts of the plane $2x - 3y + z = 12$ on the coordinate axes.
- (d) Find the coordinates of the foot of the perpendicular from the point $(1, 3, 4)$ upon the plane $2x - y + 2z = 0$.
- (e) Write the equation of straight line $x - y = 0, z = 1$ in symmetrical form.
- (f) Obtain the condition for two spheres to be orthogonal.
- (g) Find the equation of the sphere through the circle $x^2 + y^2 + z^2 = 9, 2x + 3y + 4z = 5$ and the point $(1, 2, 3)$.
- (h) If the vertex of a right circular cone be at the origin and the axis be the z -axis, then find the equation of the cone with semi-vertical angle θ .
- (i) Give the definition of reciprocal cones.
- (j) Define a right circular cylinder.
- (k) Show that five normals can be drawn to a paraboloid from a given point.

Question 2.

- (a) Find the polar equation of the normal to a conic $\frac{l}{r} = 1 - e \cos \theta$ at a point whose vectorial angle is α . [6]
- (b) If the tangents at any two points P and Q of a conic meet at a point T , and if the straight line PQ meets the directrix corresponding to focus S in a point K , then show that the angle $\angle KST$ is a right angle. [6]

Question 3.

- (a) Find the equation of the plane which passes through the points $(-1, 1, 1)$ and $(1, -1, 1)$ and is perpendicular to the plane $x + 2y + 2z = 5$. [6]

- (b) Show that the lines:

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \quad \text{and} \quad \frac{x-4}{4} = \frac{y-3}{2} = \frac{z-2}{-1}$$

are coplanar, and find the equation of the plane containing them. [6]

Question 4.

- (a) Find the coordinates of the points on the sphere $x^2 + y^2 + z^2 - 4x + 2y = 4$ where the tangent planes are parallel to the plane $2x - y + 2z = 1$. [6]

- (b) Find the equation of the sphere which passes through the two points $(0, 3, 0)$, $(-2, -1, -4)$ and cuts orthogonally the two spheres:

$$x^2 + y^2 + z^2 - 2x + 3z - 2 = 0 \quad \text{and} \quad 2(x^2 + y^2 + z^2) + x + 3y + 4 = 0$$

[6]

Question 5.

- (a) Prove that the equation $ax^2 + by^2 + cz^2 + 2ux + 2vy + 2wz + d = 0$ represents a cone if:

$$\frac{u^2}{a} + \frac{v^2}{b} + \frac{w^2}{c} = d$$

[6]

- (b) Find the condition that the plane $ux + vy + wz = 0$ may touch the cone $ax^2 + by^2 + cz^2 = 0$. [6]

Question 6.

- (a) Find the equation of the cylinder whose generators are parallel to the line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ and whose guiding curve is the ellipse $x^2 + 2y^2 = 1, z = 0$. [6]

- (b) Find the equation of a right circular cylinder of radius 2 whose axis is the line:

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{1}$$

[6]

Question 7.

- (a) A point P moves so that the section of the enveloping cone of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$, with P as vertex, by the plane $z = 0$ is a circle. Show that the locus of P is:

$$\frac{y^2(b^2 - a^2)}{b^4} + \frac{z^2(c^2 - a^2)}{c^4} = 1, \quad x = 0 \quad \text{and} \quad \frac{x^2(a^2 - b^2)}{a^4} + \frac{z^2(c^2 - b^2)}{c^4} = 1, \quad y = 0$$

[6]

- (b) Show that the plane $8x - 6y - z = 5$ touches the paraboloid $\frac{x^2}{2} - \frac{y^2}{3} = z$ and find the coordinates of the point of contact.

[6]